

INDIAN INSTITUTE OF TECHNOLOGY DELHI

New Delhi - 110016



GIPEDI INTERNSHIP REPORT

Submitted in fulfillment of the requirements for the Internship Program

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EXPERIMENT 0: ARDUINO BASIC PROGRAM

OBJECTIVE:

- Learn the components of an arduino uno circuit .
- Understand how to connect basic electronic components.
- Write and upload a simple arduino program .
- Verify the circuit through simulation .

TOOLS REQUIRED

1. Arduino UNO/esp32
2. LED
3. Arduino IDE

CIRCUIT DIAGRAM

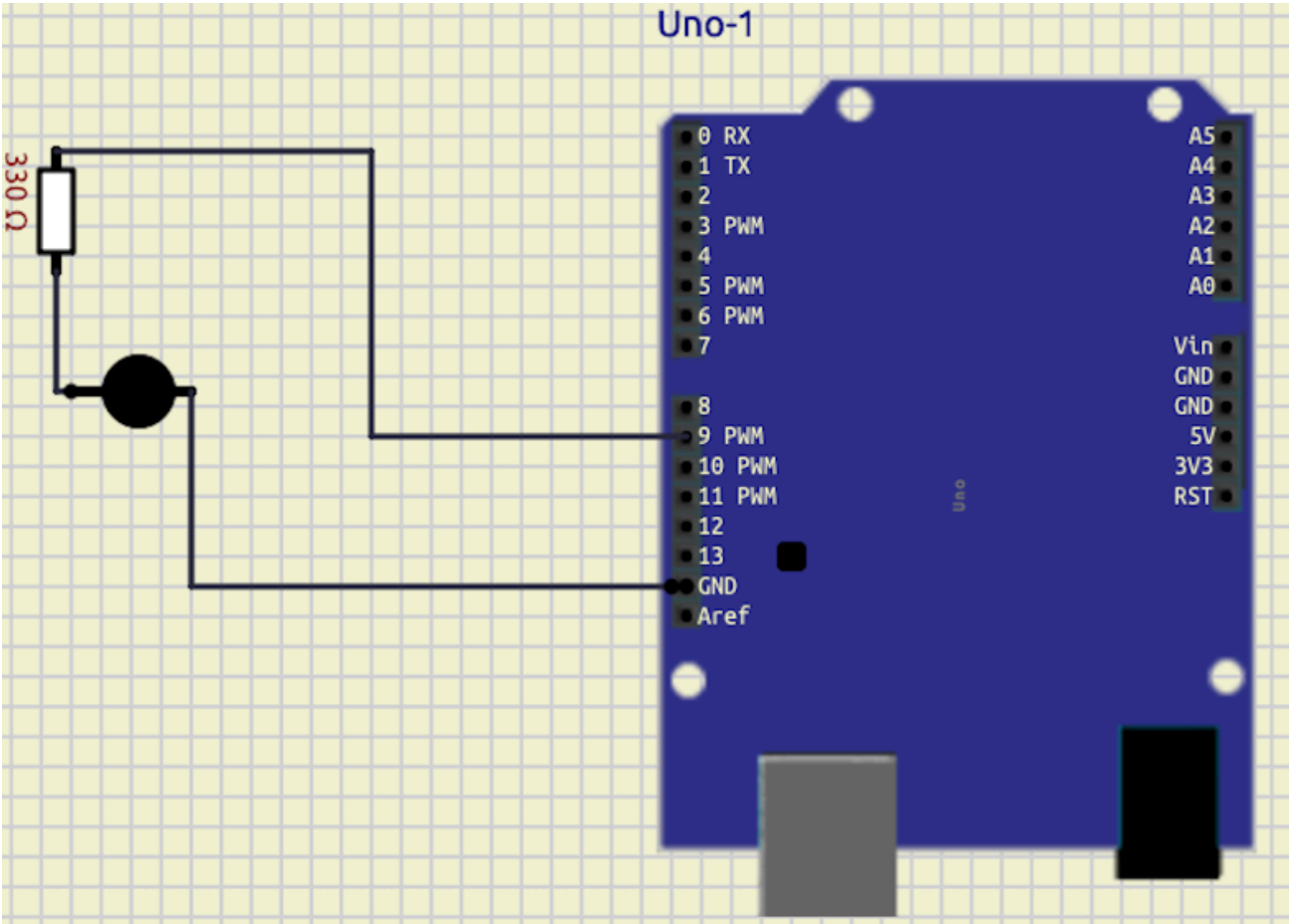


Figure 1: Inverting Amplifier Circuit

ARDUINO CODE

```
// The digital pin connected to the LED Anode
const int ledPin = 9;

void setup() {
  // Initialize the digital pin as an output
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH); // Turn the LED on
  delay(1000);                // Wait for 1 second (1000 milliseconds)
  digitalWrite(ledPin, LOW);  // Turn the LED off
  delay(1000);                // Wait for 1 second
}
```

WORKING

The program turns the LED ON for one second and OFF for one second repeatedly creating a blinking effect.

LEARNING OUTCOME

Learned how to connect an LED to arduino uno write a basic arduino program, and upload code to the board.

PROBLEMS FACED AND SOLUTIONS

- 1.Problem:LED did not blink .
- 2.solution:checked wiring connections and upload the code again.

FUTURE SCOPE

This experiment can be extended to control multiple LEDs and other electronic components.

EXPERIMENT 01 :

Interface with seven segment led to make a counter using a push button as input

OBJECTIVE

To interface a seven segment LED display with Arduino and use a push button to increment and display a counter value.

TOOLS REQUIRED

1. Arduino UNO
2. Seven segment LED display (common cathode)
3. Push button
4. Arduino IDE

THEORY

A seven segment display consists of seven LEDs arranged in the shape of the number 8 by turning ON and OFF specific segments ,digits from 0 to 9 can be displayed. A push button is used as an input device. each time the button is pressed, the counter value increases by one and the updated number is displayed on the seven -segment display.

CIRCUIT DIAGRAM

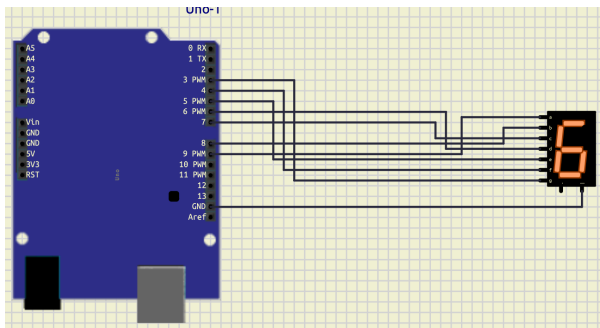


Figure 1: Interfacing with seven segment led using push button

ARDUINO CODE

```
## include <sevseg.h>
void setup (){
    //put your setup code here ,to run once :
    byte numdigits =1;
    byte digit _pins []={};
    byte seg_pins []={9,8,7,6,5,4,3,2};
    byte dis_type =COMMON_CATHODE;
    bool res_on_segs=true;
    s_seg.begin (dis_type,numdigit _pins ,seg_pins ,res_on_segs );
    s_seg.setbrightness(90);
}
void loop(){
    //put your main code here to run repeated:
    for (int i=0 ;i<10; i++)
    {
        s_seg.setnumber(i);
        s_seg.refreshdisplay ();
        delay(1000);
    }
}
```

PROCEDURE

1. Connect the seven -segment display to Arduino digital pins .
2. Connect the push button to the input pin.
3. Upload the program to arduino uno .
4. Press the push button repeatedly.
5. Observe the display number (1 to 9)

OBSERVATIONS

1. The display initially show 0.
2. Each button press increments the counter.
3. After reaching 9,the counter rests to 0.

LEARNING OUTCOME

1. Learned how to interface a seven -segment display with arduino .
2. Understood digital input using a push button .
3. Implemented a simple counter system.

PROBLEM FACED AND SOLution

PROBLEM:

Incorrect digits displayed on the seven-segment display .

SOLUTION:

Checked segment pin connections and corrected the wiring according to the circuit diagram .

FUTURE SCOPE

1. Two digital and four- digit counters.
2. Visitor counting systems.
3. Digital scoreboards.
4. Electronic voting and counting applications .

EXPERIMENT 03

Display different patterns on led matrix

OBJECTIVE

To interface an LED matrix arduino and display different patterns ,symbols and designs on the matrix .

TOOLS REQUIRED

1. Arduino uno
2. 8x8 LED matrix

THEORY

An LED matrix is a grid of LED arranged in rows and columns . by controlling the LEDs individually ,different patterns ,symbols, characters, and animations can be displayed the arduino sends data to the LED matrix to illuminate specific LEDs and create the desired pattern .

CIRCUIT DIAGRAM

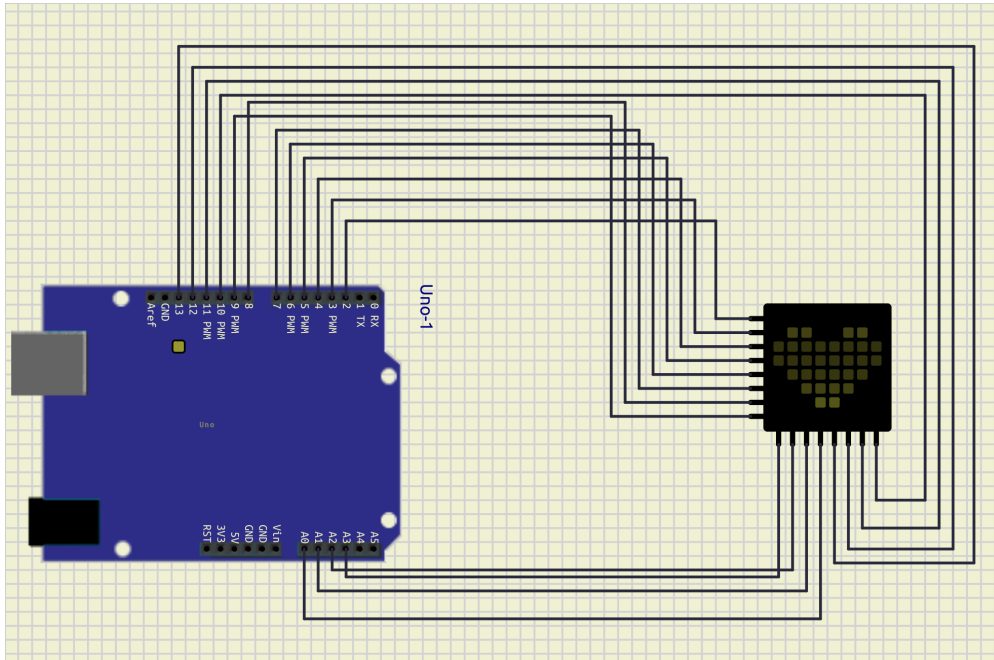


Figure To interfacing an led matrix arduino uno

ARDUINO CODE

```
int pins[] = {2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13}; // Added [] here

void setup() {
  for (int i = 0; i < 12; i++) {
    pinMode(pins[i], OUTPUT); // Fixed capital M and capitalized OUTPUT
  }
}

void loop() {
  for (int i = 0; i < 12; i++) {
    digitalWrite(pins[i], HIGH); // Fixed capital W
  }
  delay(1000); // Optional: adds a pause while they are ON

  for (int i = 0; i < 12; i++) {
    digitalWrite(pins[i], LOW); // Fixed capital W and capitalized LOW
  }
  delay(1000); // Optional: adds a pause while they are OFF
}
```

PROCEDURE

1. Connect the LED matrix to the arduino .
2. Upload the program using arduino IDE.
3. Run the circuit .
4. Observe the displayed pattern on the LED matrix .
5. Modify the pattern data to display different shapes and symbols.

OBSERVATIONS

1. The led matrix successfully displayed the programmed pattern .
2. Different patterns can be created by changing the binary values in the code .
3. The matrix can display symbols ,letters, numbers, and simple animations .

LEARNING OUTCOME

1. Learned how to interface an LED matrix with arduino
2. Understood row and columns addressing in LED matrices.
3. Gained experience in creating visual patterns using programming

PROBLEMS

Pattern was not displayed correctly .

SOLUTION {#solution } Checked wiring connections and verified the binary pattern data in the coe

.

FUTURE SCOPE

1. Display scrolling text .
2. Create animations and games
3. Design digital notice boards

PROJECT REPORT ON 555 TIME IC

TITLE

Design and implementation of a 555 timer circuit

AIM

TO study the working of the 555 timer IC and generate timing pulses using an astable multivibrator circuit

OBJECTIVE

1. To understand the operation of the 555 time IC
2. To generate a continuous square wave output
3. To observe the charging and discharging of a capacitor

COMPONENTS REQUIRED

1. 555timer IC
2. Resistor r1 (1k ohm)
3. Resistor R2(10k ohm)
capacitor c1 (10uf)
connecting wires

CIRCUIT DIAGRAM

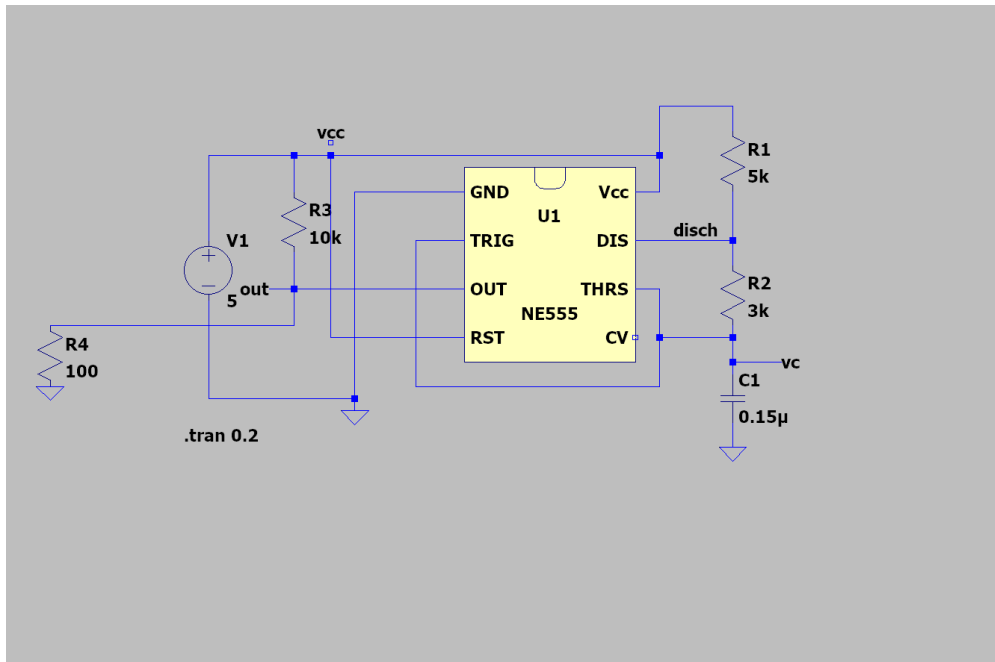


Figure 1: 555 timer IC

PROCEDURE

1. Place the timer IC on the breadboard
2. Connect the power supply.
3. Turn on the power supply
4. Observe the blinking led.

Observation

6. The LED blinking rate depends on the resistor and capacitor value
7. The LED blinks continuously
8. A square wave is obtained at the output pin .

APPLICATION

1. LED flashers
2. Pulse generators
 - clock circuit
 - alarm system

ADVANTAGES

1. Low
2. Easy to use
3. Reliable operation

CONCLUSION

The project demonstrated the working principle of the 555 timer IC the circuit successfully generated periodic pulses and showed how resistor and capacitor values affect the timing characteristic of the output signal .

INVERTING AND NON INVERTING AMPLIFIER USING OP AMP

AIM

To study and analyze the operation of inverting and non inverting amplifier using an operational amplifier (op amp)

Objective

1. To understand the working principle of inverting and non inverting amplifiers
2. To calculate and verify the voltage gain of both amplifier configurations
3. To compare their characteristics and applications .

CIRCUIT DIAGRAM INVERTING AND NON-INVERTING

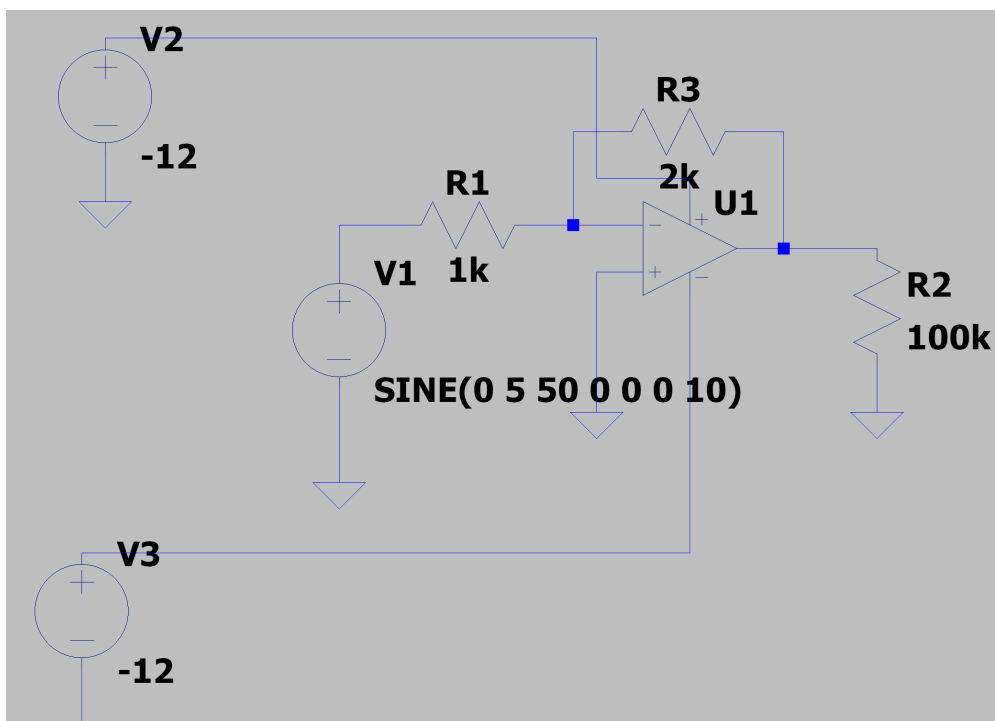


Figure 1: Inverting and non inverting amplifier using op amp

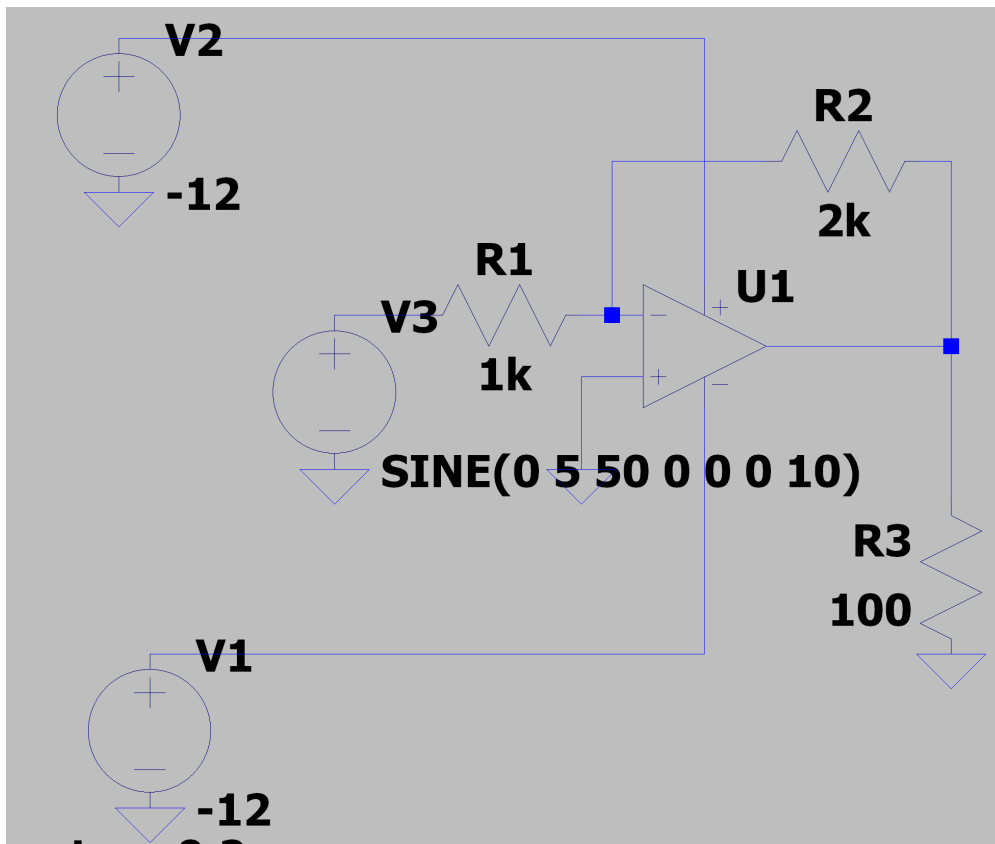


Figure 1: Inverting Amplifier Circuit

APPLICATION

INVERTING AMPLIFIER

1. Signal conditioning
2. Audio amplifiers
3. active filters

NON INVERTING AMPLIFIER

1. Voltage follows
2. Sensor signal amplification
3. Instrumentation circuits.

TYPING REPORT

sno	DATE	TYPING SCORE
1.	2/6/26	24
2.	3/6/26	24
3.	4/6/26	24
4.	5/6/26	22
5.	6/6/26	23
6.	7/6/26	26
7.	8/6/26	30
8.	9/6/26	30
9.	10/6/26	28
10	11/6/26	23
11.	12/6/26	26
12.	13/6/26	27
13.	14/6/26	27
14	15/6/26	26
15	16/6/26	26
16	17/6/26	25
17	18/6/26	26
18.	19/6/26	25
19.	20/6/26	26
20.	21/6/26	27
21.	22/6/26	27

sno	DATE	TYPING SCORE
22	23/6/26	26
23	24/6/26	26
24	25/6/26	25
25	26/6/26	26
26.	27/6/26	25
27.	28/6/26	27
28.	29/6/26	27
29	30/6/26	30
30	1 /7/26	30
31	2/7/26	29
32	3/7/26	30
33.	4/7/26	27
34	5/6/26	29
35	6/6/26	29
36	7/6/26	30
Table 1: typing report.		

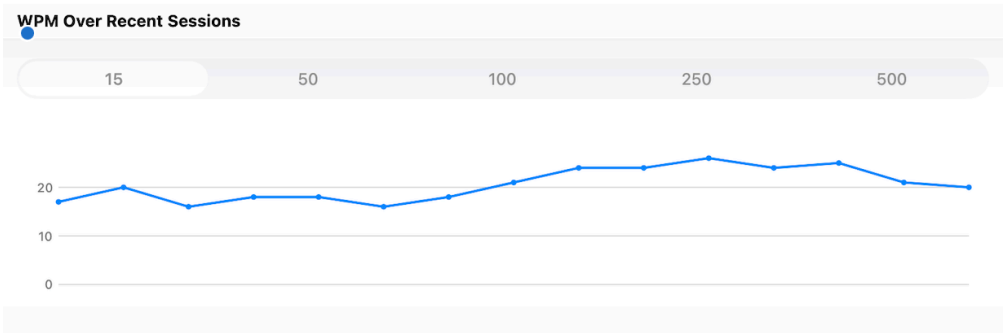


Figure Typing progress graph

HTML CODE WITH SQL DATABASE

AIM

To create a simple web page using HTML and store user information in an SQL database .

OBJECTIVE

1. To design a user input form using HTML.
2. To store the entered data in an SQL database .
3. To retrieve and display records from the database .

SOFTWARE REQUIRED

1. Notepad
2. XAMPP SERVER
3. MYsql Database
4. SQL

INTRODUCTION

HTML (hyper text markup language) is used to create web pages and collect user input through forms .SQL (Structure Query language) is used to create and manage database .since HTML cannot communicate directly with a database , a server- side language such as PHP is used as an intermediary between HTM Land SQL .this project demonstrates a simple student registration system where student can enter their details ,and the information is stored in a MYSQL database .

Software and Hardware Requirement

1. Operating system : windows 10/11
2. Text editor :NOTEPAD
3. XAMPP server
4. MySQL database

HARDWARE REQUIREMENTS

1. Computer or laptop
2. Minimum 4GB RAM
3. 500 MB free disk space

THEORY

HTML

HTML is the standard markup language used for creating web pages . it provides various elements such as forms , text boxes, button , and labels for user interaction .

SQL :

SQL is a language used to manage relational database .It allows user to create databases, table , insert record ,update data , and retrieve information .

PHP :

PHP is a server -side scripting language that processes from data and interacts with the MYSQL database

FLOW OF THE SYSTEM

1. User opens the registration page .
2. User enters details in the html form.
3. The form sends data to the php file .
4. php establishes a connection with MYSQL .
5. SQL query inserts the data into the database .
6. A success message is displayed .

SQL DATA

student_name	phy_marks	chem_marks	maths_marks	english_marks	comp_marks	total_marks	total_marks
Rahul Kumar	85	78	92	80	88	500	
Ansh Kumar	85	78	92	80	88	500	
Aaruna sharma	85	78	92	80	88	500	
Priya Sharma	90	82	95	88	91	500	
Priya Sharma	90	82	95	88	91	500	
Amit Verma	75	80	85	70	78	500	
Vinay	85	67	59	46	72	500	
Mayal	30	79	99	47	90	500	
Kavita Sharma	1	32	9	7	49	500	

Figure: student marks

HTML CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Web Page</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      background-color: #f4f4f4;
      margin-top: 50px;
    }

    h1 {
      color: #333;
    }

    button {
      padding: 10px 20px;
      font-size: 16px;
      background-color: #4CAF50;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }

    button:hover {
      background-color: #45a049;
    }
  </style>
</head>
<body>

  <h1>Welcome to My Website</h1>
  <p>This is a simple HTML page.</p>

  <button onclick="showMessage()">Click Me</button>
```

```
<script>
    function showMessage() {
        alert("Hello! Welcome to my website.");
    }
</script>

</body>
</html>
```

ADVANTAGE

1. EASY Data management
2. Reduces paperwork
3. Quick retrieval of records
4. Improves accuracy and efficiency .

APPLICATION

1. School management systems
2. College record systems
3. Employee management systems
4. Library management systems

OUTPUT

The user enters the details in the HTML form, and the information is stored in the SQL database table .

CONCLUSION

The experiment successfully demonstrated how HTML can be used to collect user data and SQL can be used to store and manage that data in a database .

BMS BUTTON USER INTERFACE

AIM

To study the operation of a push button and control an output device using a BMS (Basic Monitoring/Management System).

OBJECTIVE

To understand the working of push buttons and their application in automation and control systems.

THEORY

A Button Control is a graphical user interface (GUI) element used to execute a command when clicked by the user. It is one of the most commonly used controls in BMS applications. Button controls help users interact with software easily and efficiently.

SOFTWARE REQUIRED

1. BMS Software
2. COMPUTER SYSTEM

PROCEDURE

1. Open the BMS software
2. Create a new project.
3. Drag and drop a Button Control onto the form.
4. Change the button text to "Click Me".
5. Write code for the button click event.
6. Run the application.
7. Click the button and observe the result.

PROGRAM CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>BMS Dashboard Sketch</title>
  <style>
    body {
      font-family: 'Courier New', Courier, monospace;
      background-color: #ffffff;
      padding: 30px;
    }
    .bms-card {
      border: 2px solid #000000;
      padding: 15px;
      width: 320px;
      background: #ffffff;
      margin-bottom: 5px;
    }
    .bms-card h3 {
      margin: 0 0 10px 0;
      font-size: 18px;
      letter-spacing: 1px;
    }
    .info-text {
      margin: 5px 0;
      font-size: 15px;
    }
    .top-btns {
      margin: 10px 0;
    }
    .small-btn {
      border: 1px solid #000;
      background: none;
      padding: 2px 8px;
      margin-right: 5px;
      font-size: 12px;
    }
    .btn-section {
```

```
        margin: 15px 0;
    }
    .row {
        margin-bottom: 8px;
        display: flex;
        align-items: center;
    }
    .clickable-btn {
        border: 2px solid #000000;
        background: #ffffff;
        padding: 5px 15px;
        margin-right: 8px;
        cursor: pointer;
        font-weight: bold;
        font-family: inherit;
    }
    .clickable-btn:hover {
        background: #f0f0f0;
    }
    .arrow-label {
        font-size: 14px;
    }
    .arrow-down {
        font-size: 24px;
        margin-left: 20px;
        margin-top: -5px;
        margin-bottom: 5px;
    }
    .details-box {
        border: 2px solid #000000;
        padding: 12px;
        width: 200px;
        background: #ffffff;
        margin-left: 10px;
    }
    .details-box p {
        margin: 4px 0;
        font-size: 14px;
    }
}
</style>
</head>
<body>
```

```

<div class="bms-card">
  <h3>BMS UI</h3>
  <div class="info-text">Total Volt: 0.8V</div>
  <div class="info-text">Connection: 2S (3.2V - 4.4V)</div>

  <div class="top-btns">
    <button class="small-btn">C1</button>
    <button class="small-btn">C2</button>
  </div>x

  <div class="info-text">No. of Cells: 2</div>

  <div class="btn-section">
    <div class="row">
      <button class="clickable-btn">C1</button>
      <button class="clickable-btn">C2</button>
      <span class="arrow-label">&#x2192; clickable button</span>
    </div>
    <div class="row">
      <button class="clickable-btn">C3</button>
      <button class="clickable-btn">C4</button>
      <button class="clickable-btn">C5</button>
    </div>
    <button class="clickable-btn">C6</button>
  </div>

  </div>

  <div class="arrow-down">&#x25BC</div>

  <div class="details-box">
    <p><b id="display-name">C1</b></p>
    <p>Battery Volt: <span id="display-volt">---</span></p>
    <p>Temp: <span id="display-temp">---</span></p>
    <p>Current State: <span id="display-state">---</span></p>
  </div>

  <script>
    const dataMap = {
      'C1': { volt: '3.6V', temp: '26°C', state: 'Normal' },
      'C2': { volt: '3.4V', temp: '28°C', state: 'Normal' },
      'C3': { volt: '1.1V', temp: '45°C', state: 'Hot' },
      'C4': { volt: '0.0V', temp: '24°C', state: 'Offline' },
      'C5': { volt: '3.5V', temp: '27°C', state: 'Normal' },

```

```
    'C6': { volt: '3.3V', temp: '26°C', state: 'Normal' }  
  };
```

```
    document.querySelectorAll('.clickable-btn').forEach(btn => {  
  btn.addEventListener('click', function() {  
    const name = this.innerText.trim();  
    const info = dataMap[name];  
  
    if (info) {  
      document.getElementById('display-name').innerText = name;  
      document.getElementById('display-volt').innerText = info.volt;  
      document.getElementById('display-temp').innerText = info.temp;  
      document.getElementById('display-state').innerText = info.state;  
    }  
  });  
});  
</script>
```

SCREENSHOT

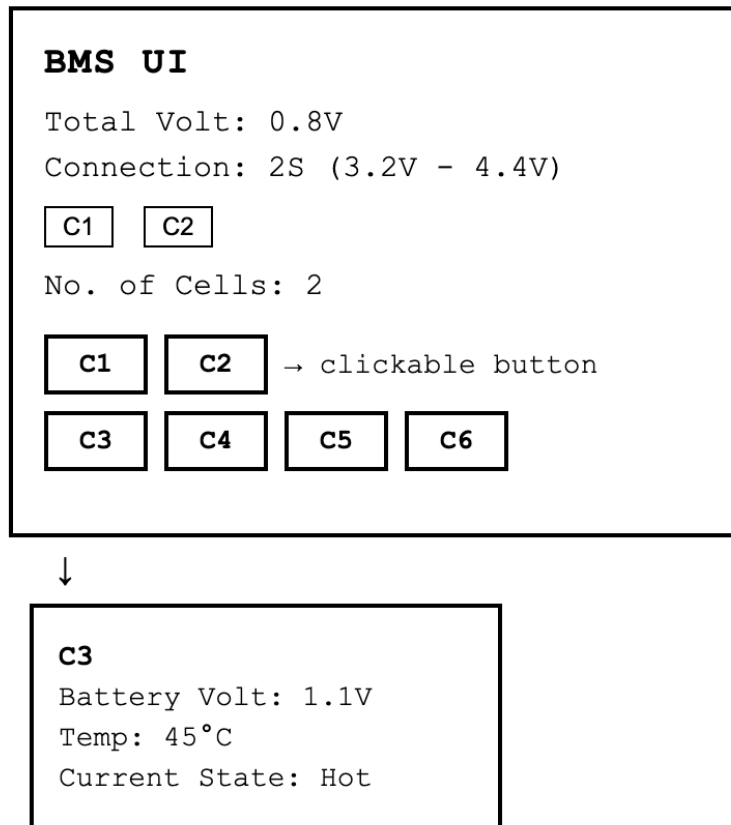


Figure bms button interface

OBSERVATION

1. The button appeared on the form.
Clicking the button
2. Displayed a message box.
3. The button executed the assigned command correctly.
Result

RESULT

The Button Control was created successfully and performed the desired action when clicked.

CONCLUSION

The Button Control is an important GUI component used to interact with applications. The experiment demonstrated how a button can be created and programmed to perform specific tasks

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